

"X – ability" Parameters
• Reliability • Availability • Maintainability • Manageability • Scalability • Mean Time to Repair (MTTR) • Mean Time Between Failures (MTBF) • Failures in Time (FIT) • Interoperability • Portability

Figure 4: 'X-ability' parameters

Software Performance Parameters
• Response Time • Latency • Data Rate • Logical Space • Frequency Band Usage • Frameworks Usage • Standards Usage • Time Loading • Memory Loading • Foot Print • Software Boot Time • Software Download Time • End to End Latency

Figure 5: Software performance parameters

measured will depend on the product in question. For example, the measure of availability of a telecom switch can be specified in five 9s. One of the performance ratings of a wireless base station can be spectrum efficiency (the

information rate or data rate that is transferred over a given bandwidth in the communication network). Different products will have different performance parameters.

The architecture we have created should satisfy the non-functional requirements or quality attributes that we have chosen in the beginning of the project, during the requirements phase. This can be measured by attaching a use case to the actual architecture decision, and describing the use case. This process is done in the Architecture Trade-off Analysis Method. Of course, all this can only be proved during the acceptance testing of the product.

Now, let us look at some of the software performance requirements.

Software performance requirements

Time loading

When we write software for a product, we add more features to satisfy customer requirements. As we add more and more features, the main CPU we use gets loaded. Time loading can be defined as the amount of useful work the CPU is doing.

In other words, the percentage of useful processing the device is doing is called time loading. Time loading is usually 75 per cent in a well-architected system, and it is supported in theory too. Time loading of 85 per cent and above indicates the processor is over-utilised, and that we should have selected a more powerful one. It is also dangerous in a real-time system. Time loading of less than 50 per cent indicates that we should have selected a lower-powered processor.

Memory loading

Memory loading is the amount of RAM used by the system when the product is functioning. The code is loaded from the flash storage to RAM during software execution, and the RAM needs to use up to 70 to 75 per cent of its capacity. Loading memory to more than 75 per cent is not advisable during execution; this fact has to be borne in mind by the architect when designing the product.

Bit error ratio

The bit error ratio is a useful parameter in communication products, and speech- and music-related products. The error ratio is the ratio of the number of incorrectly received bits, elements, characters, or blocks to the total number sent during a specified time interval. The most commonly encountered ratio is the bit error ratio (BER), sometimes also referred to as the bit error rate. Examples of the bit error ratio are:

- Transmission BER: the number of erroneous bits received, divided by the total number of bits transmitted; and (Br / Br).
- Information BER: the number of erroneous decoded (corrected) bits, divided by the total number of decoded (corrected) bits.